



MITM 431: Big Data Analytics

Lec. 1: Introduction

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- Managing data

- Relational databases/SQL
- Distributed platforms
 - Hadoop
 - Spark
- Blockchains

- Learning from data

- Descriptive/unsupervised
 - Explain the past
 - clustering, association
- Predictive/supervised
 - Predict the future (based on the past)
 - predict the value of some variable using the values of other variables



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Managing data: size

- “Small” data: spreadsheets
 - Simple data, fits in memory
- “Medium” data: relational databases
 - Database size can be larger than memory size
 - Convenient query language (SQL)
- “Big” data:
 - Performance optimizations
 - Distributed platforms: Hadoop, Spark
 - But SQL remains a popular query language

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Managing data: trust

- Traditional applications: trusted party maintains data
 - E.g., banks maintain transaction ledgers
- What if we remove the third party?
 - E.g., cryptocurrencies
 - How to manage data in a decentralized and tamper-evident manner?
 - Solution: blockchains



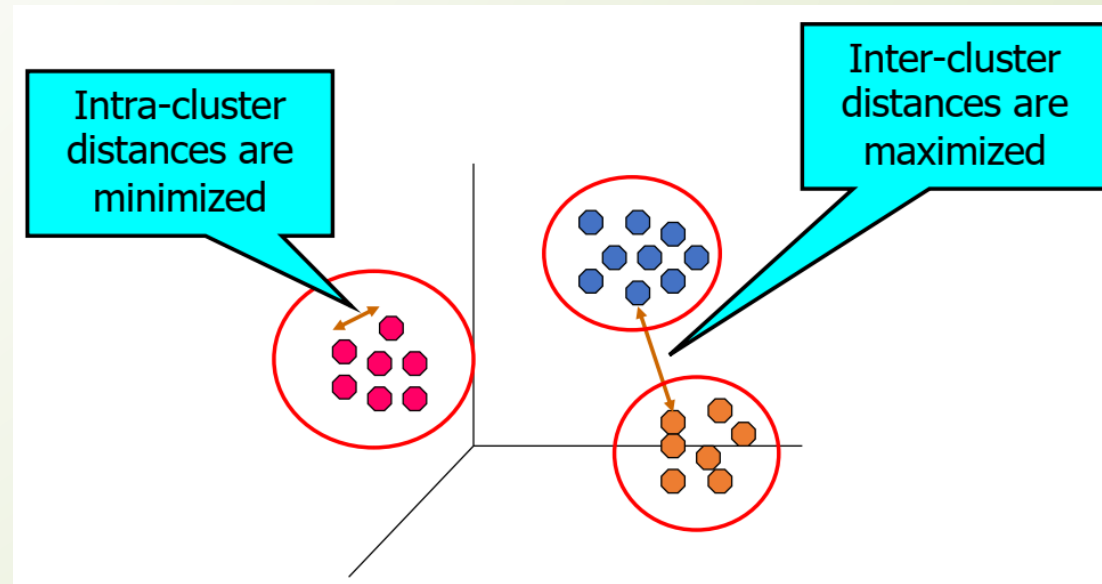
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Clustering

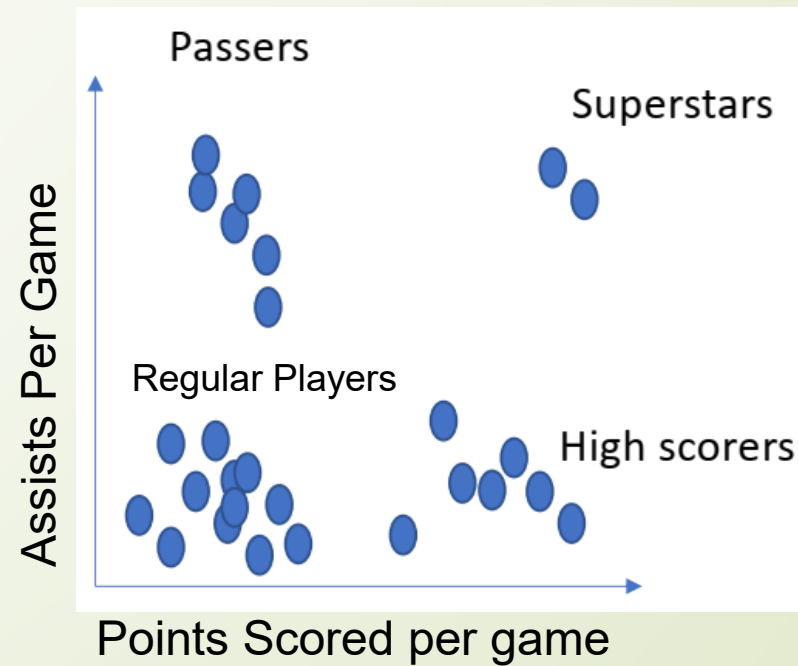
Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



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Clustering (contd.)

Example: basketball player statistics



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Association

- Given a set of records, each of which contains some number of items from a given collection,
- produce dependency rules to predict the occurrence of an item based on occurrences of other items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

$\{\text{Milk}\} \rightarrow \{\text{Coke}\}$

$\{\text{Diaper, Milk}\} \rightarrow \{\text{Beer}\}$

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Association: Business Insight

- Suppose we've discovered $A \rightarrow B$
 - We can consider putting A and B on the same shelf
 - If sales of A increase, sales of B may also increase
 - If we discontinue selling A, maybe the sales of B will drop



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Predictive analytics

- Supervised learning – predict the future
- To predict a numeric variable:
 - Least-squares linear regression
- To predict a discrete/categorical variable:
 - Classification: logistic regression, nearest neighbor classifier, decision trees, Naïve Bayes, etc.



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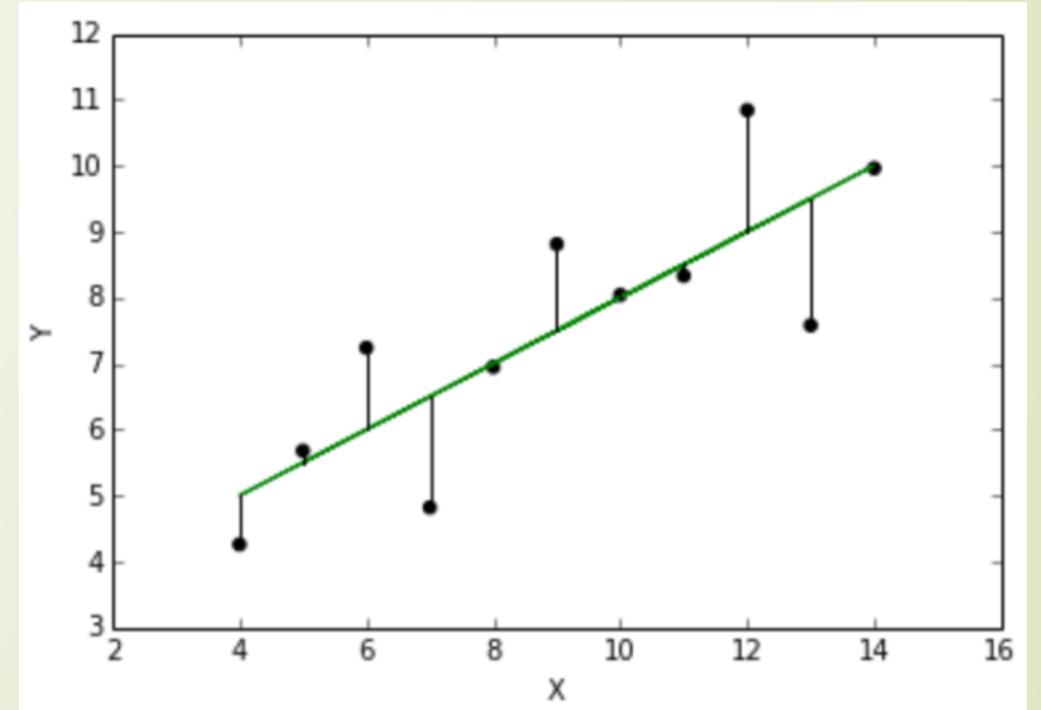
Predictive analytics

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Least-squares linear regression

- $y = m_1x_1 + m_2x_2 + m_3x_3 + \dots + b$
- Supervised: we are given a dataset with the values of the x_i 's and the values of y
- Residual = actual y – predicted y
- Optimization goal: minimize residual sum of squares



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Least-squares linear regression

Example: movie revenue prediction

- Data source:
 - Internet movie data base (IMDB)
- Simple model:
 - $\text{Movie revenue} = m_1 * (\text{number of A-list actors}) + m_2 * (\text{number of B-list actors}) + b$
- Improved model:
 - $\text{Movie revenue} = m_1 * (\text{number of A-list actors}) + m_2 * (\text{number of B-list actors}) + m_3 * (\text{is_sequel}) + b$
- “Feature engineering” is important!
- Association Mining: if{Jashim} then {Shabana}



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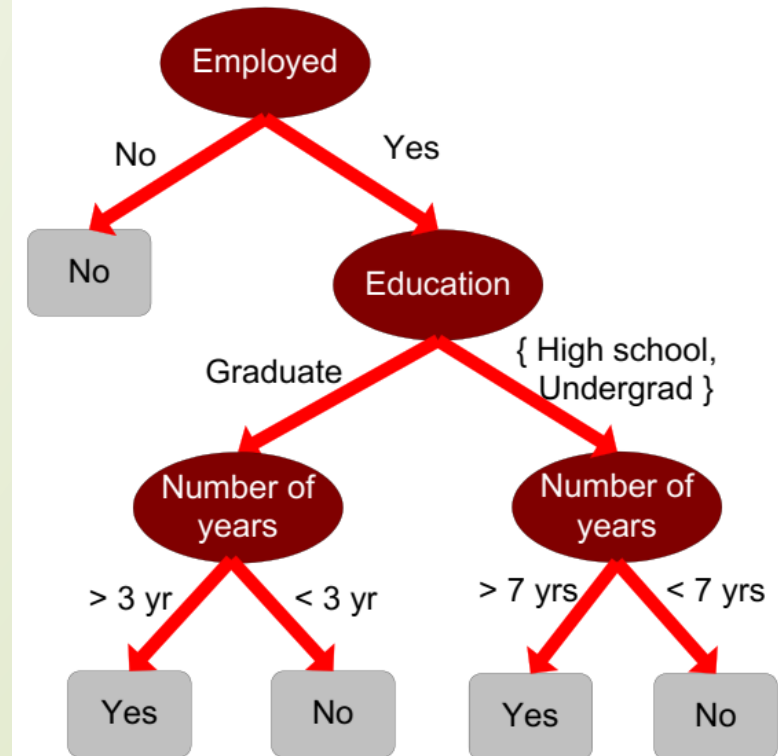
Example: decision tree classifier

- Supervised learning: we are given the values of the class variable

Class

<i>Tid</i>	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
...	...	...	...	...

Model for predicting credit worthiness



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Hands On:

Which engineering program is right for you?

- See: http://www.engineering.uwaterloo.ca/undergraduate/quiz/quiz_html_js.html
- Classification:
 - Collect data from students (your program, are you happy in your program?, answers to survey questions)
 - Remove unhappy students
 - Learn a classifier to predict program based on answers to survey questions
- Association:
 - if "question 1 = a" then "question 5 = b"



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For more information

- See Textbook Chapter 1
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